DUNMAN SECONDARY SCHOOL

CANDIDATE NAME			
CLASS		INDEX NUMBER	

PRELIMINARY EXAMINATION 2024
SECONDARY 4 EXPRESS/ 5 NORMAL ACADEMIC

MATHEMATICS **4052/01**
Paper 1

2 hours 15 minutes

Marking Scheme

Question	Answer
1a	27^4 $=(3^3)^4$ $=3^{12}$
1b	$5^x=5^6+5^6+5^6+5^6+5^6$ $5^x=5(5^6)$ $5^x=5^7$ $x=7$

Question	Answer
2	$54=2\times3^2$ $78=2\times3\times13$ $204=2^2\times3\times17$
	HCF = $2\times3=6$
	Min Cubes $=\frac{54}{6}\times\frac{78}{6}\times\frac{204}{6}$ $=3978$

Question	Answer
3	$a=2$ $b=5$ $c=2$

Question	Answer
4	$a=\dfrac{4+3c^2}{c^2-2b}$ $ac^2-2ab=4+3c^2$ $ac^2-3c^2=4+2ab$ $c^2(a-3)=4+2ab$ $c^2=\dfrac{4+2ab}{a-3}$ $c=\pm\sqrt{\dfrac{4+2ab}{a-3}}$

Question	Answer
5	Big Square Perimeter : Cross Perimeter $2:1$ $24:12$ Big Square Perimeter : Small Square Perimeter $24:4$ $6:1$ Big Square Area : Small Square Area $6^2:1^2$ $36:1$ Big Square Area : Cross Area 36 :5

Question	Answer
6	$\frac{4x}{3}-\frac{5x-1}{2}=-3$ $\frac{8x-15x+3}{6}=-3$ $-7x+3=-18$ $-7x=-21$ $x=3$

Question	Answer
7	Total amount after 7 years $=5000(1+\frac{4}{100})^7$ $=6579.66$ Remaining amount after withdrawal = 3579.66 Since <u>**new Principal amount is less than initial Principal**</u> amount, interest earned is the same duration will be lesser. <u>**Tammy is incorrect.**</u> Or Finding and compare interests earned between two durations <u>**$1579.66 vs $1130.93**</u> <u>**Tammy is incorrect.**</u>

Question	**Answer**
8a	$7(3y+2x)-2(x-4y)$ $=21y+14x-2x+8y$ $=29y+12x$
8b	$12ax-6xy-4by+8ab$ $=12ax+8ab-6xy-4by$ $=2[6ax+4ab-3xy-2by]$ $=2[2a(3x+2b)-y(3x+2b)]$ $=2(2a-y)(3x+2b)$

Question	Answer
9a	LQ $=\dfrac{95+98}{2}$ $=96.5$ UQ $=\dfrac{X+129}{2}$ $IQR=UQ-LQ$ $30=\dfrac{X+129}{2}-96.5$ $126.5=\dfrac{X+129}{2}$ $253=X+129$ $X=124$ $x=4$
9b	$117.5=\dfrac{Y+118}{2}$ $235=Y+118$ $Y=117$ $y=7$

9c	On average, the girls are taller than the boys as the girls' median (117.5 cm) is higher than the boys' median (112 cm). Or There are more girls that are more than 120 cm as to compare to the boys

Question	Answer
10	Volume of hemisphere $= \frac{2}{3}\pi(3r)^3$ $= 18\pi r^3$ Volume of cylinder $= \pi(2r)^2 h$ $= 4\pi r^2 h$ $18\pi r^3 = 2(4\pi r^2 h)$ $18r = 8h$ $9r = 4h$ $h = \frac{9r}{4}$ Total surface area of cylinder $= 2\pi(2r)^2 + 2\pi(2r)(\frac{9r}{4})$ $= 8\pi r^2 + 9\pi r^2$ $= 17\pi r^2$

Question	Answer
11a	P(all three late) $=(0.15)^3$ $=\dfrac{27}{8000}$
11b	P(at least one depart on time) $=1-\dfrac{27}{8000}$ $=\dfrac{7973}{8000}$
11c	P(exactly 2 out of 3 ferries depart on time) = P(only 1 is late) $=(0.15)(0.85)(0.85)\times 3$ $=\dfrac{2601}{8000}$

Question	Answer
12a	Vertical axis does not starts from zero.
12b	Reader may perceive that profit increases close to 500% from 2019 to 2023, instead of actual 9% to 10%.

Question	Answer
13a	Perpendicular bisector drawn correctly
13b	Angle bisector drawn correctly
13c	Point X marked correctly

Question	Answer
14	$p = kq^3$ where k is a constant $24 = kq^3$ $k = \dfrac{24}{q^3}$ Let $q_1 = \dfrac{1}{2}q$ $p_1 = k(q_1)^3$ $p_1 = \dfrac{24}{q^3}\left(\dfrac{1}{2}q\right)^3$ $p_1 = 3$

Question	Answer
15a	$\frac{15}{t} = 0.5$ $t = 30$
15b	$750 = \frac{1}{2}[90 + (t - 30)](15)$ $100 = 60 + t$ $t = 40$ length of time $= 40 - 30 = 10$ Or $750 = \frac{1}{2}[90 + T](15)$ $100 = 90 + T$ $T = 10$

Question	Answer
16a	$x^2 + 7x + 10$ $= (x + 3.5)^2 + 10 - 3.5^2$ $= (x + 3.5)^2 - 2.25$
16b	$(-3.5, -2.25)$

Question	Answer
17a	Factors of 16
17b	3, 5, 7, 11, 13
17c	$B \cup A'$

Question	Answer
18a	045°
18b	7.9 cm (measured from the map) t$=\dfrac{7.9\times665}{730}$ $=7.1965$ h =7 hour 12 min

Question	Answer
19a	$p=7$ $q=23$ $r=31$
19b	$8n-1$
19c	$8n-1=3012$ $8n=3013$ $n=376.625$ Since n is not an integer, 2012 is not a term of this sequence.

Question	Answer
20a	$PQ=\sqrt{(4+9)^2+(-3-1)^2}$ $=\sqrt{185}$ $=13.6$ units
20b	Gradient $=\dfrac{1-(-3)}{-9-4}$ $=-\dfrac{4}{13}$ Using $m=-\dfrac{4}{13}$ and $(4,-3)$, $-3=-\dfrac{4}{13}(4)+c$ $-3=-\dfrac{16}{13}+c$ $c=-\dfrac{23}{13}$ Equation of PQ: $y=-\dfrac{4}{13}x-\dfrac{23}{13}$

Question	**Answer**
21a	$AC^2 + AB^2$ $= 3.2^2 + 6^2$ $= 46.24$ $BC^2 = 6.8^2$ $= 46.24$ Since $AC^2 + AB^2 = BC^2$, By Converse of Pythagoras' Theorem, Triangle *ABC* is a right angle triangle with angle *CAB* being a right angle where *BC* is the longest side. Thus, *AB* is perpendicular to *AC*. (Shown)
21b	$\cos \angle BCD = -\frac{6}{6.8} = -\frac{15}{17}$ $\sin \angle ABC = \frac{6}{6.8} = \frac{15}{17}$ 0

Question	Answer
22a	Rhombus
22b	Ext Angle $= \dfrac{360°}{5}$ $= 72°$ or Total Int Angle $= (5-2)\times 180°$ $= 540°$ Int Angle $= 180° - 72°$ $= 108°$ or Int Angle $= \dfrac{540°}{5}$ $= 108°$
22c	Reflex angle $= 360° - 108°$ $= 252°$
22d	$\angle YZX$ $= \dfrac{360° - 2\times 108°}{2}$ $= 72°$ $\angle AZY$ $= 108° - 72°$ $= 36°$

Question	Answer
23a	$\cos 55° = \dfrac{OS}{4.5}$ $OS = 4.5\cos 55°$ $= 2.5810$ $= 2.58$ m
23b	$TS = \sqrt{4.5^2 - 2.5810^2}$ $= 3.6862$ $\tan \angle TXS = \dfrac{3.6862}{2.5810 + 3}$ $\angle TXS = 33.4°$ Angle $= 33.4° + 35°$ $= 68.4°$

Question	Answer
24ai	$\overrightarrow{RP} = -\underset{\sim}{r} + \underset{\sim}{p}$
24aii	$\overrightarrow{OA} = \frac{1}{2}\underset{\sim}{r} + \frac{1}{2}\underset{\sim}{p}$
24aiii	$\overrightarrow{QS} = -1.5\underset{\sim}{p} + 3\underset{\sim}{r}$
24bi	$\overrightarrow{OB}$ $= \overrightarrow{OQ} + \overrightarrow{QB}$ $= 1.5\underset{\sim}{p} + \frac{1}{3}(-1.5\underset{\sim}{p} + 3\underset{\sim}{r})$ $= \underset{\sim}{r} + \underset{\sim}{p}$
24bii	Since $\overrightarrow{OA} = \frac{1}{2}\overrightarrow{OB}$, O, A and B are collinear. OA is half the length of OB.
24biii	Area of $\triangle APO = 12$ Area of $\triangle BPA = 12$ (Using common height for BPO with BO as base) Area of $\triangle BPO = 24$ Area of $\triangle BPQ = \frac{24}{2} \times 1 = 12$ (Using common height for BQO with QO as base) Area of $\triangle BOQ = 12 + 24 = 36$